

# ROBERT VOKÁČ, BSc

## C++ Systems Engineer (Runtime / Compatibility Layers)

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GitHub: [github.com/robertvokac](https://github.com/robertvokac) · [github.com/openeggbert](https://github.com/openeggbert)

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## PROFILE

C++ engineer focused on **low-level systems, runtime design, and API compatibility layers**.

Designed and implemented **custom runtimes replacing XNA, DirectX 3, and WinAPI**, enabling legacy applications and games to run without original dependencies.

Experience includes **reverse-engineering behavior, reproducing runtime semantics, and building cross-platform abstractions over SDL3/OpenGL**.

Transitioning from Java backend development to **full-time C++ systems engineering**, with focus on runtime and compatibility layers.

Seeking roles in **systems engineering, runtime/framework development, game technology, or backend/platform architecture**.

## CORE PROJECTS

### CNA

GitHub: [github.com/openeggbert/cna](https://github.com/openeggbert/cna)

**Custom C++ runtime reimplementing XNA 4.0 API over SDL3 (~5k LOC, production-tested on real game)**

- Designed **compatibility layer matching XNA runtime behavior** (game loop, input, rendering pipeline)
- Implemented **rendering abstraction** supporting SDL\_Renderer, OpenGL, and bgfx backends
- Built **resource lifecycle, input/audio systems, and framework architecture**
- Ported real-world game (**Speedy Blupi, 2013**) from C# / XNA to C++
- Reproduced original runtime behavior **without access to original engine internals**
- Achieved **smooth runtime performance** (~0.1% CPU usage, ~47 MB RAM)

**Playable web build:** <https://speedyblupi.com/SpeedyBlupi2013/>

**Video demo:** <https://www.youtube.com/watch?v=TScL1L-dbHE>

### Free Direct + Free API

GitHub: [github.com/openeggbert/free-direct](https://github.com/openeggbert/free-direct) · [github.com/openeggbert/free-api](https://github.com/openeggbert/free-api)

**Custom reimplementation of DirectX 3 (2D) and WinAPI subset in modern C++ over SDL3 (~3k LOC per module)**

- Implemented **DirectDraw-style 2D rendering and WinAPI-style event/input handling**.
- Built compatibility layer enabling execution of **legacy Windows applications without original dependencies**.
- Investigating and resolving memory management issues in long-running scenarios

- Supports execution of:
  - **Speedy Blupi (1998/2001)** without DirectX SDK or WinAPI
  - **Planet Blupi (1996)** (partial compatibility, ongoing debugging)

Playable web build: <https://speedyblupi.com/SpeedyEggbert2/>

Video demo: <https://www.youtube.com/watch?v=TI3vNrpgtbo>

## Hive (archived)

GitHub: [github.com/robertvokac/hive](https://github.com/robertvokac/hive)

40k+ LOC modular backend platform

- Designed **metadata-driven architecture, custom ORM, and REST generation**
- Implemented **plugin system, scheduler, triggers, and validation pipeline**

## SKILLS

**Languages:** C++20/23 (primary), SQL

**Systems & Tools:** SDL3, OpenGL, CMake, Linux, SQLite, Git

### Core Areas:

- Runtime and framework architecture
- API compatibility layers (XNA / DirectX / WinAPI)
- Rendering and system integration
- Modular systems and abstraction design
- Debugging complex systems and legacy behavior

## EXPERIENCE

### Software Engineer – Assist (2017 - present)

- Maintained and improved **production backend systems (Java, SQL)**
- Diagnosed and resolved **production issues, improving system stability**
- Applied **system design principles in long-lived codebases**
- Developed **C++ systems independently (runtime and compatibility layers)**

Technologies: Java, SQL, Git, Bash

## EDUCATION

### BSc in Computer Science

Czech University of Life Sciences Prague (2014-2017)

## LANGUAGES

English - B2

Czech - native